

Benefiting From Single-Objective Feature Selection to Multiobjective Feature Selection: A Multiform Approach

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Abstract—Evolutionary multiobjective feature selection (FS) has gained increasing attention in recent years. However, it still faces some challenges, for example, the frequently appeared duplicated solutions in either the search space or the objective space lead to the diversity loss of the population, and the huge search space results in the low search efficiency of the algorithm. Minimizing the number of selected features and maximizing the classification performance are two major objectives in FS. Usually, the fitness function of a single-objective FS problem linearly aggregates these two objectives through a weighted sum method. Given a predefined direction (weight) vector, the single-objective FS task can explore the specified direction or area extensively. Different direction vectors result in different search directions in the objective space. Motivated by this, this article proposes a multiform framework, which solves a multiobjective FS task combined with its auxiliary single-objective FS tasks in a multitask environment. By setting different direction vectors, promising feature subsets from single-objective FS tasks can be utilized, to boost the evolutionary search of the multiobjective FS task. By comparing with five classical and state-of-the-art multiobjective evolutionary algorithms, as well as four well-performing FS algorithms, the effectiveness and efficiency of the proposed method are verified via extensive experiments on 18 classification datasets. Furthermore, the effectiveness of the proposed method is also investigated in a noisy environment.

Index Terms—Classification, evolutionary feature selection (FS), multiform optimization, multiobjective optimization.

I. INTRODUCTION

THE MASSIVE data today poses great challenges for data mining. Classification is a crucial task in data mining,

which intends to categorize a set of data into classes based on the information provided by its features. However, a large number of features used for data analysis can lead to the issue of “the curse of dimensionality.” Since not all available features are informative, the existence of irrelevant, noisy, and redundant features can significantly degrade the performance of a classifier and increase the training time [1], [2]. Feature selection (FS) is one of the most important tasks in data preprocessing for data mining, which aims to select a small subset of representative features from the original feature set that has strong discrimination ability over a given dataset [3], [4], [5]. The benefits of FS for classification are manifold, such as saving computation time to increase computational efficiency, building a more generalized classification model to avoid overfitting, removing redundant and irrelevant features to boost the classification performance, and saving the original semantics of features to provide a more interpretable classification model [4].

The current mainstream of FS algorithms can be broadly categorized into three categories [6]: 1) filter; 2) wrapper; and 3) embedded. The filter-based methods evaluate a feature subset on the basis of the intrinsic characteristics of the data that are independent of a classifier’s performance. The wrapper-based methods evaluate the importance of a feature subset according to the performance of a classifier. The embedded-based methods learn a classifier and perform FS simultaneously. Usually, the wrapper-based methods are more time consuming than the other two methods since evaluating each feature subset requires training a classifier. However, they generally obtain better classification results since the classification performance is directly used for feature subsets evaluation. Thus, this article focuses on developing a wrapper-based approach for FS.

Maximizing the classification performance and reducing the feature subset size are two major objectives for FS, which can be formulated as a multiobjective optimization problem. Evolutionary algorithms (EAs) have been widely employed to address multiobjective optimization problems due to their numerous advantages, such as global search capability, less requirement for domain knowledge, and population-based search to obtain multiple solutions. Such derived algorithms are typically called multiobjective EAs (MOEAs) [7]. Due to the nature of MOEAs, a set of nondominated solutions can be found in a single run, and these tradeoff solutions, that is, feature subsets, can meet different requirements

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of decision makers in real-world applications. Evolutionary multiobjective FS has gained increasing attention in recent years [6], [8].

Pareto dominance-based and decomposition-based algorithms are two commonly used MOEAs [7]. The Pareto dominance-based MOEAs usually analyze the domination relationship between solutions and rank them to different fronts through nondominated sorting, which can provide enough convergence pressure toward the Pareto front for bi- and tri-objective optimization problems. Despite their advantages, they still face some limitations when solving multiobjective FS problems. For instance, they usually focus on the central of the Pareto front [9], which cannot easily find feature subsets with high classification performance or a small number of selected features. Furthermore, their diversity maintenance is ineffective for multiobjective FS problems due to the frequently appeared duplicated solutions [10]. Decomposition-based algorithms decompose a multiobjective optimization problem into a set of single-objective subproblems by a set of predefined direction (weight) vectors and solve them in parallel. One key issue in decomposition-based algorithms, however, is the distribution of the obtained Pareto-optimal solutions is highly dependent on the setting of direction vectors. It is very challenging for decomposition-based algorithms to specify a set of appropriate direction vectors for multiobjective FS problems, mainly because a priori information about the geometry of the Pareto front is unknown, which is different from many of the current benchmarks in the evolutionary multiobjective community, where most of them are designed for numeric and continuous benchmark optimization problems.

Single-objective FS usually considers the classification performance and the selected feature subset size via a weighted sum method [11], [12], [13], [14]. It uses a specific direction vector to control the proportion of these two terms. The direction vector can be seen as a representative of a subregion in the objective space. Given a direction vector, the single-objective FS algorithm can explore a specified direction or area extensively.

Multiform optimization is an emerging paradigm which exploits alternate formulations of a given optimization task [15], [16], [17], [18]. Since different formulations of the given optimization problem can capture different properties of the landscapes of the problem encountered, useful traits obtained from these alternate formulations through evolutionary searches can enhance the problem-solving capability of the original optimization task.

The overall goal of this article is to develop an MOEA to find a diverse set of tradeoff feature subsets between the classification performance and the number of selected features, by synthesizing both single-objective FS and multiobjective FS, which can make full use of their strengths and effectively avoid their weaknesses. To achieve this goal, this article proposes a multiform optimization framework to solve multiobjective FS problems, called MFFS. MFFS manipulates three evolutionary subpopulations in a multitask setup, and each subpopulation is associated with a specific form of FS problem. The main population derives from the original multiobjective FS task

to be solved, which focuses on searching for a set of non-dominated solutions. The other two auxiliary subpopulations originate from the constructed single-objective FS tasks. By periodically adjusting the direction vectors in single-objective FS tasks, they aim to first achieve the fast convergence along two boundary direction vectors, to find more feature subsets with better classification performance and a smaller number of selected features, and then compensate the possible diversity loss of the population in the multiobjective FS task by exploring some new regions in the objective space.

The main contributions of this article can be summarized as follows.

- 1) A multiform framework is proposed, which constructs two simplified single-objective FS problems to serve as auxiliary tasks for the original multiobjective FS task. The useful information offered by each task can be utilized through continuous knowledge exchange, to improve the performance of the multiobjective FS task.
- 2) A search direction adjustment mechanism is proposed. It first forces the search toward the two boundary directions to find more feature subsets with better classification performance and a smaller number of selected features, and then adjusts the direction vectors of single-objective FS tasks according to the solutions' distribution of the multiobjective FS task, which can compensate the possible diversity loss of the main population by exploring some new regions in the objective space.
- 3) A novel diversity maintenance mechanism is proposed. By considering the similarity and distribution of feature subsets in both the objective and search spaces, the proposed method can filter out the unpromising duplicated feature subsets, thereby improving the population diversity and retaining the potentially optimal feature subsets on the test data to improve generalization.

The remainder of this article is organized as follows. Section II reviews the related work about multiobjective FS. The details of the proposed MFFS algorithm are elaborated in Section III. Section IV presents the experiment design. Section V reports the experimental results. Section VI provides some further analysis. Finally, Section VII concludes this article.

II. RELATED WORK

Since most of the existing MOEA-based FS methods can be classified as dominance-based and decomposition-based, this section reviews some related work on these two approaches.

A. Pareto Dominance-Based MOEAs for FS

In the past decade, various Pareto dominance-based multiobjective FS algorithms have been suggested. The pioneering work using multiobjective particle swarm optimization (PSO) for FS is presented in [19], where PSO with the Pareto dominance, crowding, and mutation is used to search for a set of tradeoff feature subsets. To improve the quality of feature subsets obtained by the multiobjective PSO algorithms, an adaptive uniform mutation strategy and a local learning strategy are designed in [20], which can effectively improve

the exploration and exploitation capabilities of the population, respectively. In FS, some features can discriminate all class labels, while some of them can only distinguish a part of class labels. In order to identify these two kinds of features simultaneously, a multiobjective memetic algorithm [21] is proposed for multiclass cancer classification.

To discover the hidden relationship between feature subsets and class labels in binary classification problems, a filter-based multiobjective genetic programming [22] is proposed to find solutions with minimum cardinality and maximum relevance. To perform FS and classification simultaneously, a multiobjective genetic programming [23] is developed. It divides a multiclass FS problem into multiple binary classification problems, and evolves multiple genetic programs to build ensembles. Later, an improved version of multiobjective genetic programming [24] is proposed for FS and feature extraction. In this method, each binary classifier consists of a binary tree and a support vector machine (SVM). The extracted features from the tree are used to train the SVM, and the SVM determines the outcome of the corresponding classifier.

An improved Pareto dominance-based MOEA [10] is proposed to analyze and handle the duplicated feature subsets in both the search and objective spaces. In this algorithm, a duplication analysis is used to filter out the unpromising feature subsets based on their similarity between feature subsets. To improve the search efficiency and enhance the solution quality on high-dimensional datasets, the steering matrix [25] is suggested to reflect the importance of each feature to the solution and guide the search to promising areas.

Different features could be acquired at different costs, and the cost-sensitive FS problem can be viewed as a multiobjective optimization problem. To deal with such problems, the two-archive MOEA [26] is developed to improve the population diversity in both the objective and search spaces through the external and leader archives, respectively. In some cases, the cost of obtaining a feature is fuzzy. A fuzzy dominance relationship and a fuzzy crowding distance [27] are developed, which extends NSGA-II [28] to the context of fuzzy objectives, and the results show that they could obtain feature subsets with superior performance in terms of diversity, approximation, and feature cost. Datasets with missing data are most likely to be unreliable. To select features with high reliability, a new multiobjective FS formulation [29] is established which considers the missing rate as an objective. Similarly, a multipopulation differential evolution algorithm with a dynamic special crowding distance mechanism [30] is introduced, enabling the algorithm to find multiple feature subsets that have the same objective values but select different features.

The preferences for the two objectives in FS are not different. It is well-recognized that classification performance is more important than the number of selected features. To solve such unbalanced objective preference problems, a fitness function combining the classification performance and Pareto dominance principle [31] is used to put more emphasis on classification performance. In [32], a multiobjective FS problem is reformulated as a constrained multiobjective

FS problem, which imposes a constraint on the classification performance based on the distribution of nondominated solutions, to preferentially select feature subsets with better classification performance.

Due to the need to remove redundant and irrelevant features, multiobjective FS can be regarded as a kind of sparse multiobjective optimization problem. Some specific techniques have been developed to ensure its sparsity, for example, the estimated score of each feature [33] and the sparse distribution learning based on pattern mining [34].

B. Decomposition-Based MOEAs for FS

There are also some decomposition-based MOEAs for multiobjective FS. To execute instance selection and FS simultaneously, a genetic algorithm is applied to optimize the classification performance and the number of selected features and instances via a weighted sum function [35], to search for a small number of feature and instance subsets that can provide high generalization ability to the classifier.

In FS, classification accuracy usually takes precedence over the number of selected features, and they do not always conflict with each other. To handle such issues, a reference-based MOEA/D [9] is suggested, which uses reference points instead of direction vectors to pay more attention to feature subsets with higher classification accuracy. Recently, combined with gradient descent local search, the reference-based MOEA/D is applied for sparsity-based multilabel FS [36].

C. Discussions

From the above reviews, we can find that Pareto dominance-based algorithms are the most popular methods for multiobjective FS. This can be attributed to the nondominated sorting operator employed in most dominance-based MOEAs can easily identify the dominance relationship between feature subsets, and select nondominated solutions to provide convergence pressure toward the Pareto front. They also require no extra parameters and less computational resources. Moreover, their shortcoming of the deterioration of the selection pressure in many-objective space does not exist in bi-objective FS.

However, the Pareto dominance-based MOEAs still face limitations when solving multiobjective FS problems. For example, they usually focus on the central of the Pareto front [9], which cannot easily find feature subsets with high classification performance or a small number of selected features. Furthermore, their diversity maintenance is ineffective for solving multiobjective FS problems [10]. This is because if a dataset has a small number of training instances or a large number of redundant features, duplicated solutions will frequently appear in multiobjective FS problems. Selection of all these duplicated feature subsets in the objective space can greatly degenerate the diversity and lead to the premature convergence of solutions, due to the overlap of these duplicated solutions in the objective space. Last but not least, on high-dimensional FS, the canonical MOEAs inevitably exhibit low search efficiency due to the huge search space.

In view of the limitations of previous work, in the next section, we will propose a multiform framework to address

the above issues. Specifically, to address the first issue, we use two subpopulations to search along the two boundary directions [37] to overcome the shortcoming that Pareto dominance-based MOEAs only focus on the central section of the Pareto front, to find more feature subsets with better classification performance and a smaller number of selected features. Then, the direction vectors are adjusted to search in sparse areas to compensate for the diversity loss of the multiobjective FS task. In addition, to handle the frequently appeared duplication of solutions in either the search space or the objective space, a diversity maintenance mechanism is proposed to filter out the unpromising duplicated solutions by considering the similarity of solutions in both the objective space and the search space.

III. PROPOSED ALGORITHM

In this section, first, two different forms of FS tasks are defined. Then, an overview of the proposed MFFS method is presented. After that, the two main proposed components of MFFS: 1) the search direction adjustment mechanism and 2) the diversity maintenance mechanism, are introduced in detail.

A. Definition of Two Forms of FS Tasks

1) *Multiobjective FS Task*: Minimizing the selected feature ratio $f_{\text{ratio}}(\mathbf{x})$ and the classification error rate $f_{\text{err}}(\mathbf{x})$ can be considered as a multiobjective optimization problem

$$\begin{aligned} \min \mathbf{F}(\mathbf{x}) &= (f_{\text{ratio}}(\mathbf{x}), f_{\text{err}}(\mathbf{x})) \\ \text{where } \mathbf{x} &= (x_1, \dots, x_D) \in \Omega = \{0, 1\}^D \end{aligned} \quad (1)$$

where D is the total number of features in a dataset. $\mathbf{x}$ is a solution that represents a feature subset. In this article, binary encoding is used since it is more straightforward than a continuous encoding in FS [38], [39], where $x_i = 1$ and $x_i = 0$ mean that the i th feature is selected and discarded, respectively.

$f_{\text{ratio}}(\mathbf{x})$ equals the ratio of the number of selected features to the total number of features

$$f_{\text{ratio}}(\mathbf{x}) = \frac{\sum_{i=1}^D x_i}{D} \quad (2)$$

$f_{\text{err}}(\mathbf{x})$ is obtained by evaluating a certain classifier using the selected feature subset $\mathbf{x}$. Since many collected data are unbalanced, a balanced classification error rate [40] is used

$$f_{\text{err}}(\mathbf{x}) = 1 - \frac{1}{c} \sum_{i=1}^c \frac{TP_i}{|S_i|} \quad (3)$$

where c is the number of classes of a dataset. TP_i represents the number of correctly identified instances in class i , and $|S_i|$ denotes the sample size of class i .

2) *Single-Objective FS Task*: The widely used fitness function in single-objective FS [11], [12], [13] is a combination of $f_{\text{err}}(\mathbf{x})$ and $f_{\text{ratio}}(\mathbf{x})$

$$\begin{aligned} \min F(\mathbf{x}) &= (1 - \lambda) \times f_{\text{ratio}}(\mathbf{x}) + \lambda \times f_{\text{err}}(\mathbf{x}) \\ \text{where } \mathbf{x} &= (x_1, \dots, x_D) \in \Omega = \{0, 1\}^D \end{aligned} \quad (4)$$

where $\lambda \in [0, 1]$ is the weight to control the contribution of the classification error rate and the selected feature ratio.

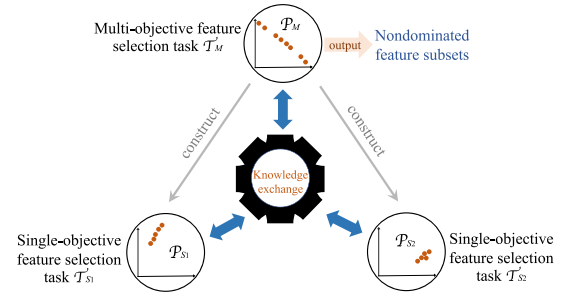


Fig. 1. Multi-form framework for multiobjective FS.

Actually, by setting different λ values, (4) is similar to the widely used weighted sum aggregate function in MOEA/D [41] for multiobjective optimization problems

$$\begin{aligned} \min F^{ws}(\mathbf{x}|\boldsymbol{\lambda}, \mathbf{z}^*) &= \sum_{i=1}^m (\lambda_i |f_i(\mathbf{x}) - z_i^*|) \\ \text{where } \mathbf{x} &= (x_1, \dots, x_D) \in \Omega = \{0, 1\}^D \end{aligned} \quad (5)$$

where m is the number of objectives ($m = 2$ in this article). $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_m)$ is the direction vector and $\sum_{i=1}^m \lambda_i = 1$. $\mathbf{z}^* = (z_1^*, \dots, z_m^*)$ is the ideal point specified by the best value of each objective among evaluated solutions.

The proposed MFFS algorithm considers two different kinds of tasks: 1) a multiobjective FS task $\mathcal{T}_M$ in (1) and 2) two single-objective FS tasks $\mathcal{T}_{S1}$ and $\mathcal{T}_{S2}$ in (5). $\mathcal{T}_M$, $\mathcal{T}_{S1}$, and $\mathcal{T}_{S2}$ all belong to the same domain $\mathcal{D} = \{\Omega, p(\mathbf{x})\}$, where Ω represents the search space and $p(\mathbf{x})$ is the probability distribution of feature subsets. In other words, in the proposed multitask setup, the target (multiobjective FS) task $\mathcal{T}_M$ and the auxiliary (single-objective FS) tasks $\mathcal{T}_{S1}$ and $\mathcal{T}_{S2}$ have the same search space Ω , as they share the same set of features and underlying probability distribution $p(\mathbf{x})$ of solutions.

Fig. 1 illustrates the relationship between $\mathcal{T}_{S1}$, $\mathcal{T}_{S2}$, and $\mathcal{T}_M$. As illustrated in Fig. 1, MFFS conducts an evolutionary search on three FS tasks, each corresponding to a particular (minimization) optimization problem, and all tasks are evolved in parallel. We focus on solving the multiobjective FS task $\mathcal{T}_M$ assisted by two auxiliary single-objective FS tasks $\mathcal{T}_{S1}$ and $\mathcal{T}_{S2}$ via a multitask environment. During the optimization process, different tasks can assist each other by sharing knowledge with others. When we optimize these two forms of tasks separately, it is reasonable to assume that due to the same probability distribution of solutions and the relationship between $\mathcal{T}_{S1}$, $\mathcal{T}_{S2}$, and $\mathcal{T}_M$, the single-objective FS task can serve as an auxiliary, to improve the extensivity and compensate the diversity loss of the approximated Pareto front obtained by the multiobjective FS task $\mathcal{T}_M$, through setting the proper direction vectors.

B. Overall MFFS Framework

Algorithm 1 presents the overall framework of MFFS. MFFS divides computational resources into two equal parts for multiobjective FS and single-objective FS tasks, respectively. In this way, assuming the total population size is N , the population size N_M for the multiobjective FS task $\mathcal{T}_M$ is

Algorithm 1: MFFS

Input: N : population size.
Output: the nondominated solutions in $\mathcal{P}_M$.

- 1 Initialize $\lambda_{S_1} = (0.99, 0.01)$ and $\lambda_{S_2} = (0.01, 0.99)$;
- 2 $\mathcal{P} \leftarrow$ Initialize N solutions from the search space Ω ;
- 3 $\mathcal{P}_M \leftarrow$ Select N_M solutions from $\mathcal{P}$ based on *Algorithm 3*;
- 4 $\mathcal{P}_{S_1} \leftarrow$ Select N_S solutions from $\mathcal{P}$ based on $F^{ws}(\mathbf{x}|\lambda_{S_1}, \mathbf{z}^*)$;
- 5 $\mathcal{P}_{S_2} \leftarrow$ Select N_S solutions from $\mathcal{P}$ based on $F^{ws}(\mathbf{x}|\lambda_{S_2}, \mathbf{z}^*)$;
- 6 **while** the termination criterion is not met **do**
- 7 $\mathcal{O}_M \leftarrow$ Offspring generation for $\mathcal{T}_M$;
- 8 $\mathcal{O}_{S_1} \leftarrow$ Offspring generation for $\mathcal{T}_{S_1}$;
- 9 $\mathcal{O}_{S_2} \leftarrow$ Offspring generation for $\mathcal{T}_{S_2}$;
- 10 Evaluate the objective values of $\mathcal{O}_M$, $\mathcal{O}_{S_1}$, and $\mathcal{O}_{S_2}$, respectively;
- 11 Update the ideal point $\mathbf{z}^*$;
- 12 $\mathcal{P}_M \leftarrow$ Environmental selection from $\mathcal{P}_M \cup \mathcal{O}_M \cup \mathcal{O}_{S_1} \cup \mathcal{O}_{S_2}$; */* Algorithm 3 */*
- 13 $\mathcal{P}_{S_1} \leftarrow$ Environmental selection from $\mathcal{P}_{S_1} \cup \mathcal{O}_M \cup \mathcal{O}_{S_1} \cup \mathcal{O}_{S_2}$ based on $F^{ws}(\mathbf{x}|\lambda_{S_1}, \mathbf{z}^*)$;
- 14 $\mathcal{P}_{S_2} \leftarrow$ Environmental selection from $\mathcal{P}_{S_2} \cup \mathcal{O}_M \cup \mathcal{O}_{S_1} \cup \mathcal{O}_{S_2}$ based on $F^{ws}(\mathbf{x}|\lambda_{S_2}, \mathbf{z}^*)$;
- 15 **if** $\mathcal{P}_{S_1}$ and $\mathcal{P}_{S_2}$ are not improved in consecutive l generations **then**
- 16 Update λ_{S_1} and λ_{S_2} for $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$, respectively; */* Algorithm 2 */*
- 17 Reinitialize $\mathcal{P}_{S_1}$ and $\mathcal{P}_{S_2}$.
- 18 **Return** the nondominated solutions of $\mathcal{P}_M$.

$N/2$ and the population size N_S for two single-objective FS tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$ is $N/4$, that is, $N = N_M + 2 \times N_S$.

For initialization, N solutions are randomly generated from the entire search space Ω , where the number of features selected in each solution is randomly produced between 1 and D . Note that we limit the number of selected features to be no more than three times the population size, that is, $|\mathbf{x}| = \text{rand}(1, \min(D, 3N))$, to facilitate convergence for high-dimensional datasets [10]. The tasks $\mathcal{T}_M$, $\mathcal{T}_{S_1}$, and $\mathcal{T}_{S_2}$ select N_M , N_S and N_S solutions from them to form their initial populations $\mathcal{P}_M$, $\mathcal{P}_{S_1}$, and $\mathcal{P}_{S_2}$, respectively.

At each generation, populations $\mathcal{P}_M$, $\mathcal{P}_{S_1}$, and $\mathcal{P}_{S_2}$ will correspondingly generate an offspring population $\mathcal{O}_M$, $\mathcal{O}_{S_1}$, and $\mathcal{O}_{S_2}$, respectively. Information exchange between different tasks occurs in the environment selection process, that is, for each FS task, the selection of elite solutions as the next parent population happens not only from the parent and offspring populations of its own task, but also considers the offspring solutions of other tasks (lines 12–14 of Algorithm 1). In this way, good solutions between tasks can be transferred and utilized to improve search efficiency.

Given the direction vectors λ_{S_1} and λ_{S_2} , the environmental selection operations of tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$ are based on the fitness values of $F^{ws}(\mathbf{x}|\lambda_{S_1}, \mathbf{z}^*)$ and $F^{ws}(\mathbf{x}|\lambda_{S_2}, \mathbf{z}^*)$ in (5),

respectively. For task $\mathcal{T}_M$, the combination of the parent and the offspring populations can contain many duplicated solutions, partly because different FS tasks might generate the same offspring solutions, and also because FS is a kind of combinatorial optimization problem. If these duplicated solutions are not analyzed and filtered, the performance of the algorithm will degenerate dramatically [10]. To deal with this issue, after performing the nondominated sorting operator, MFFS has an additional diversity maintenance mechanism to handle duplicated solutions in both the search space and the objective space, which will be described in Section III-D.

C. Search Direction Adjustment

In order to search along two boundary directions, we set the initial direction vectors in (5) to $\lambda_{S_1} = (0.99, 0.01)$ and $\lambda_{S_2} = (0.01, 0.99)$ for single-objective FS tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$, respectively. The reason for this setting rather than $\lambda_{S_1} = (1, 0)$ and $\lambda_{S_2} = (0, 1)$ is that, once feature subsets have the same number of selected features or classification error rates, the feature subset with less classification error rate or a smaller number of selected features will be preferred, which can select highly relevant features and remove redundant features.

The single-objective FS tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$ are used to extensively exploit the specific directions or areas defined by direction vectors λ_{S_1} and λ_{S_2} until no improvement can be made. If the best objective value for each single-objective FS task is not improved in l successive generations, they might be stuck into local optima, and more computational resources should be reallocated to directions or areas that have not been fully explored by $\mathcal{T}_M$. Thus, the direction vectors λ_{S_1} and λ_{S_2} should be adjusted, which is achieved by reinitializing the subpopulations of $\mathcal{P}_{S_1}$ and $\mathcal{P}_{S_2}$ (lines 15–17 of Algorithm 1).

In multiobjective optimization, uniformly distributed Pareto optimal solutions can be achieved by adding subproblems in sparse regions [42]. The search direction adjustment for single-objective FS tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$ is based on the distribution of nondominated solutions in $\mathcal{T}_M$. An example of the search direction adjustment is shown in Fig. 2. In Fig. 2, we first find the two nondominated objective vectors in the objective space with the largest angle representing sparse areas, that is, $\mathbf{F}(\mathbf{x}_1)$ and $\mathbf{F}(\mathbf{x}_2)$, and connect them as a segment. After that, the partition point (i.e., the midpoint $\mathbf{p}_1$ in Fig. 2) on the segment is found. Finally, we can draw another line connecting the partition point $\mathbf{p}_1$ and the ideal point $\mathbf{z}^*$. As a result, this line is the new direction vector λ_{S_1} for task $\mathcal{T}_{S_1}$. In the same way, another direction vector λ_{S_2} for task $\mathcal{T}_{S_2}$ can be obtained. Algorithm 2 presents the details of calculating the new search directions.

The angle between two adjacent nondominated objective vectors is calculated as follows:

$$\theta(\mathbf{F}(\mathbf{x}_1), \mathbf{F}(\mathbf{x}_2)) = \arccos\left(\frac{(\mathbf{F}(\mathbf{x}_1) - \mathbf{z}^*) \bullet (\mathbf{F}(\mathbf{x}_2) - \mathbf{z}^*)}{\|\mathbf{F}(\mathbf{x}_1) - \mathbf{z}^*\| \times \|\mathbf{F}(\mathbf{x}_2) - \mathbf{z}^*\|}\right) \quad (6)$$

where $\mathbf{F}(\mathbf{x}) = (f_{\text{ratio}}(\mathbf{x}), f_{\text{err}}(\mathbf{x}))$. $(\mathbf{F}(\mathbf{x}_1) - \mathbf{z}^*) \bullet (\mathbf{F}(\mathbf{x}_2) - \mathbf{z}^*)$ represents the inner product of two adjacent nondominated objective vectors $\mathbf{F}(\mathbf{x}_1)$ and $\mathbf{F}(\mathbf{x}_2)$. $\|\cdot\|$ calculates the norm of a direction. It is clear that the angle between $\mathbf{F}(\mathbf{x}_1)$ and $\mathbf{F}(\mathbf{x}_2)$ is in the range of $[0, (\pi/2)]$.

Algorithm 2: Direction Vector Adjustment**Input:** $\mathcal{P}_M$: the parent population of task $\mathcal{T}_M$.**Output:** λ_{S_1} and λ_{S_2} .

- 1 $z_i^* = \min_{\mathbf{x} \in \mathcal{P}_M \cup \mathcal{P}_{S_1} \cup \mathcal{P}_{S_2}} f_i(\mathbf{x})$, $i = 1, 2$;
- 2 Calculate the angle of each of two adjacent nondominated objective vectors of $\mathcal{P}_M$ according to Eq. (6);
- 3 **for** $i = 1 : 2$ **do**
- 4 Find the two objective vectors that have the largest angle, and reset the angle value to 0;
- 5 Calculate the midpoint of these two objective vectors;
- 6 Calculate the direction vector λ_{S_i} of the midpoint according to Eq. (9);
- 7 **Return** λ_{S_1} and λ_{S_2} .

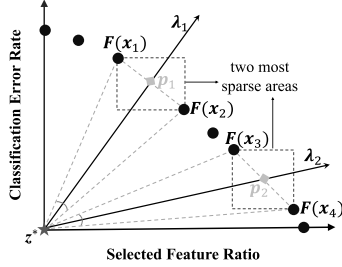


Fig. 2. Schematic for search direction (weight vector) adjustment.

Once obtaining the two solutions (e.g., $\mathbf{x}_1$ and $\mathbf{x}_2$) with the largest objective angle, we can calculate their midpoint $\mathbf{x}_p$

$$\begin{aligned} & (f_{\text{ratio}}(\mathbf{x}_p), f_{\text{err}}(\mathbf{x}_p)) \\ &= \left(\frac{f_{\text{ratio}}(\mathbf{x}_1) + f_{\text{ratio}}(\mathbf{x}_2)}{2}, \frac{f_{\text{err}}(\mathbf{x}_1) + f_{\text{err}}(\mathbf{x}_2)}{2} \right). \end{aligned} \quad (7)$$

Assume $\mathbf{z}^* = (z_1^*, z_2^*)$ is the ideal point, and $\lambda = (\lambda, 1 - \lambda)$ is the direction vector for $\mathbf{x}_p$. Then, it holds that

$$\frac{f_{\text{ratio}}(\mathbf{x}_p) - z_1^*}{\lambda} = \frac{f_{\text{err}}(\mathbf{x}_p) - z_2^*}{1 - \lambda}. \quad (8)$$

Based on (8), it is easy to deduce that

$$\lambda = (\lambda, 1 - \lambda) = \left(\frac{f_{\text{ratio}}(\mathbf{x}_p) - z_1^*}{f_{\text{ratio}}(\mathbf{x}_p) - z_1^* + f_{\text{err}}(\mathbf{x}_p) - z_2^*}, \frac{f_{\text{err}}(\mathbf{x}_p) - z_2^*}{f_{\text{ratio}}(\mathbf{x}_p) - z_1^* + f_{\text{err}}(\mathbf{x}_p) - z_2^*} \right). \quad (9)$$

Note that nondominated solutions may not be found in the most sparse area. To avoid repeating the search in directions/regions that have already been searched, if the current midpoint of the most sparse area is equal to or very close to the midpoint of the previous state, we ignore this sparse area and find another one.

Once the new search directions λ_{S_1} and λ_{S_2} are determined, the populations $\mathcal{P}_{S_1}$ and $\mathcal{P}_{S_2}$ need to be reinitialized to search along the new directions. To avoid the new populations starting the search from scratch, the current population $\mathcal{P}_M$ of $\mathcal{T}_M$ is used to generate new solutions in the reinitialization step. We flip two bits for each solution in $\mathcal{P}_M$ which is similar to the

bit-flip mutation operation. Specifically, we randomly flip one bit from 1 to 0, and flip another bit from 0 to 1. The benefit of this method is that it can use the prior knowledge learned in $\mathcal{P}_M$, and the number of selected features will not be changed, which can ensure the sparsity of new solutions.

It is worth noting that, unlike most existing direction vector adjustment methods which are normally designed to directly seek uniformly distributed solutions for numeric continuous multiobjective optimization problems with irregular Pareto fronts [43], the proposed direction vector adjustment method just plays an auxiliary role to alleviate the issue that the Pareto dominance-based MOEAs usually focus on the central of the Pareto front, which cannot easily find feature subsets with high classification performance or a small number of selected features, and also to compensate the diversity loss of the multiobjective FS task.

D. Diversity Maintenance by Handling Duplicated Feature Subsets

The duplication of feature subsets in the search and objective spaces can lead to poor diversity and premature convergence of the population. They should be handled differently.

1) *Handling Duplicated Feature Subsets in the Search Space:* The offspring population is likely to contain duplicated solutions in the search space, that is, the newly generated feature subset selects the same features as its parent population. This situation occurs more frequently in tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$, since the optima of the single-objective task is one solution. Evaluating and reserving these duplicated feature subsets in the search space wastes the computational cost and leads to diversity loss. To keep the number of solutions in the population unchanged, we repair these duplicated solutions in the search space before evaluating them. Similarly, we randomly flip one bit from 0 to 1 and one bit from 1 to 0, respectively. By flipping two bits for each duplicated solution, the first objective value, that is, f_{ratio} , will not be changed.

2) *Handling Duplicated Feature Subsets in the Objective Space:* The duplicated solutions in the objective space are more challenging to deal with than those in the search space. Intuitively, the duplicated solutions in the objective space with low similarity to the selected solutions should be reserved, since they might choose relevant features that are more different from those selected by the chosen solutions, and can provide more diversified information for the population.

Algorithm 3 presents the pseudo-code of the environmental selection operator for task $\mathcal{T}_M$, which adds additional steps to handle duplicated solutions in the objective space. After performing the nondominated sorting for the combined population $\mathcal{P}_M \cup \mathcal{O}_M \cup \mathcal{O}_{S_1} \cup \mathcal{O}_{S_2}$, the nondominated solutions will be saved directly to the next parent population (line 5 of Algorithm 3), regardless of whether they are duplicated in the objective space. This is because they represent the best feature subsets found so far.

Lines 8–23 of Algorithm 3 present the details of dealing with dominated duplicated solutions. For dominated duplicated solutions in the objective space, retaining them without filtering will deteriorate the diversity due to their overlap in the

Algorithm 3: Environmental Selection of $\mathcal{T}_M$

Input: $\mathcal{P}_M$: the parent population of task $\mathcal{T}_M$.
 $\mathcal{O}_M$, $\mathcal{O}_{S_1}$, and $\mathcal{O}_{S_2}$: the offspring populations of tasks $\mathcal{T}_M$, $\mathcal{T}_{S_1}$, and $\mathcal{T}_{S_2}$, respectively.
 N_M : population size of task $\mathcal{T}_M$.
Output: the parent population $\mathcal{P}_M$.

```

1  $aDist \leftarrow$  Calculate the average distance of solutions in
   $\mathcal{P}_M$  in the search space;
2  $\mathcal{P} \leftarrow \mathcal{P}_M \cup \mathcal{O}_M \cup \mathcal{O}_{S_1} \cup \mathcal{O}_{S_2}$ ;
3 Remove duplicated solutions in the search space of  $\mathcal{P}$ ;
4  $(\mathcal{F}_1, \mathcal{F}_2, \dots) \leftarrow$  Perform nondominated sorting on  $\mathcal{P}$ ;
5  $\mathcal{P}_M \leftarrow \mathcal{F}_1$ ;
6  $i \leftarrow 2$ ;
7 while  $|\mathcal{P}_M| < N_M$  do
8   for  $\mathbf{x} \in |\mathcal{F}_i|$  do
9     if  $\mathbf{x}$  is unique in the objective space then
10        $\mathcal{P}_M \leftarrow \mathcal{P}_M \cup \mathbf{x}$ ;
11     else
12        $\mathcal{R} \leftarrow \{\mathbf{x}' | f_{ratio}(\mathbf{x}') == f_{ratio}(\mathbf{x}), \mathbf{x}' \in \mathcal{P}_M\}$ ;
13       if  $|\mathcal{R}| == 0$  then
14          $\mathcal{R} \leftarrow \{\mathbf{x}' | \arg \min_{\mathbf{x}' \in \mathcal{F}_1} (|f_{ratio}(\mathbf{x}') - f_{ratio}(\mathbf{x})|)\}$ ;
15        $d(\mathbf{x}, \mathcal{R}) \leftarrow$  Calculate the distance between  $\mathbf{x}$ 
        and  $\mathcal{R}$ ;
16       if  $|d(\mathbf{x}, \mathcal{R}) \geq aDist| \geq 1$  then
17          $S_1 \leftarrow \{\mathbf{x} | d(\mathbf{x}, \mathcal{R}) \geq aDist\}$ ;
18       else
19          $S_1 \leftarrow \{\mathbf{x} | \arg \max_{\mathbf{x}} d(\mathbf{x}, \mathcal{R})\}$ ;
20        $\mathcal{P}_M \leftarrow \mathcal{P}_M \cup S_1$ ;
21     if  $|\mathcal{P}_M| \geq N_M$  then
22       break
23    $i \leftarrow i + 1$ ;
24 Truncate the size of  $\mathcal{P}_M$  to  $N_M$  by crowding distance;
25 Return  $\mathcal{P}_M$ .
```

objective space. We consider the distance between these duplicated solutions and their nearest reference solution set $\mathcal{R}$ in the search space, attempting to enable duplicated solutions with large dissimilarity to $\mathcal{R}$ to survive. In this article, the Hamming distance is used to measure the similarity between two feature subsets in the search space. The determination of the nearest reference solution set $\mathcal{R}$ for each duplicated solution in the objective space has the following two cases.

- 1) If the duplicated solutions in the objective space have the same number of selected features as some solutions in the selected solution set $\mathcal{P}_M$, for example, $\mathbf{x}_3$ in Fig. 3(a), these solutions will be saved as $\mathcal{R}$ (line 12 of Algorithm 3).
- 2) If both the duplicated solutions and the solutions in $\mathcal{P}_M$ have different selected feature ratios, the non-dominated solution with the smallest difference in the selected feature ratio from these duplicated solutions is saved as $\mathcal{R}$, for example, $\mathbf{x}_3$ in Fig. 3(b) (line 14 of Algorithm 3).

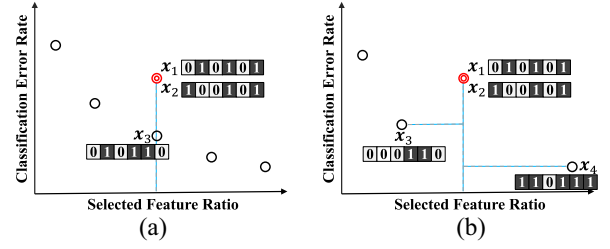


Fig. 3. Schematic of determining the reference solution set $\mathcal{R}$ (i.e., $\mathbf{x}_3$) for duplicated solutions in the objective space ($\mathbf{x}_1$ and $\mathbf{x}_2$). (a) Duplicated solutions have the same number of selected features as the selected solution set. (b) Duplicated solutions have the different number of selected features as the selected solution set.

The solutions in $\mathcal{R}$ have the same or very similar number of selected features but smaller classification error rates than those duplicated solutions. Unless these duplicated solutions can select very different features from the solutions in $\mathcal{R}$, they should be removed to maintain the population diversity.

Once the reference solution set $\mathcal{R}$ of these duplicated solutions is determined, we can decide which duplicated solutions should be deleted or saved according to their similarity to $\mathcal{R}$. Specifically, if the distance between a duplicated solution and $\mathcal{R}$ is larger than the average distance $aDist$ of the parent population $\mathcal{P}_M$, which suggests that this solution has good diversity in the search space, so it will be reserved to $\mathcal{P}_M$ (line 17 of Algorithm 3). Otherwise, only the duplicated solution with the largest distance from $\mathcal{R}$ is survived to $\mathcal{P}_M$ to maintain the diversity of the objective space (line 19 of Algorithm 3).

When the size of $\mathcal{P}_M$ equals or exceeds N_M , the above selection process stops. After that, the crowding distance operator is utilized to truncate the size of the population $\mathcal{P}_M$ to N_M (line 24 of Algorithm 3).

E. Analysis of Computational Complexity

The computational cost of MFFS comes from both two kinds of tasks. For the two single-objective FS tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$, the worst-case time complexities of the mating selection, genetic operators, and environmental selection are $2 \times O(N_S) = O(N/2)$, $2 \times O(DN_S) = O(ND/2)$, and $2 \times O(N \log N)$, respectively, where D is the total number of features, N_S is the population size in the single-objective FS task, and N is the total population size. For the multiobjective FS task $\mathcal{T}_M$, the worst-case time complexities of the mating selection, genetic operators, and environmental selection are $O(N_M) = O(N/2)$, $O(DN_M) = O(ND/2)$, and $O(N^2)$, respectively, where N_M is the population size in the multiobjective FS task.

Therefore, the overall complexity of one generation of MFFS is bounded by $\max\{O(DN), O(N^2)\}$, which is similar to its baseline NSGA-II. However, the running speed of MFFS is much faster than NSGA-II, since it can select a small number of features. This will be verified in Section VI-C.

IV. EXPERIMENT DESIGN

A. Classification Datasets

A total of 18 real-world classification datasets (Table I in the supplementary material) [44], [45] are utilized for testing

TABLE I
COMPARATIVE RESULTS OF HV VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY NSGA-II,
MOEA/D, MOEA/D-AWA, CA-MOEA, DEA-GNG, AND MFFS ON THE TEST DATA

Dataset	NSGA-II [28]	MOEA/D [41]	MOEA/D-AWA [42]	CA-MOEA [46]	DEA-GNG [47]	MFFS
Leaf	5.6918E-1(1.69E-2)+	5.8794E-1(2.50E-2)+	6.0198E-1(1.57E-2)+	5.8517E-1(2.29E-2)+	6.0937E-1(1.16E-2)+	6.2347E-1(6.38E-3)
German	6.8660E-1(1.39E-2)+	6.6570E-1(2.81E-2)+	6.8206E-1(1.61E-2)+	6.7161E-1(2.25E-2)+	6.8164E-1(1.36E-2)+	6.9179E-1(1.13E-4)
WBCD	9.3823E-1(7.33E-3)+	9.2569E-1(8.63E-3)+	9.3232E-1(7.09E-3)+	9.3231E-1(1.29E-2)+	9.3524E-1(8.87E-3)+	9.4549E-1(2.38E-3)
Dermatology	8.9116E-1(5.80E-3)≈	8.5816E-1(1.34E-2)+	8.8757E-1(7.42E-3)≈	8.8504E-1(8.38E-3)+	8.8718E-1(6.18E-3)≈	8.9082E-1(4.55E-3)
Ionosphere	8.8187E-1(2.08E-2)+	8.7861E-1(2.08E-2)+	8.8966E-1(1.35E-2)+	8.8175E-1(1.62E-2)+	8.8907E-1(1.34E-2)+	9.0299E-1(7.62E-3)
Sonar	8.2707E-1(2.96E-2)≈	8.1891E-1(3.17E-2)≈	8.4741E-1(2.60E-2)≈	8.2712E-1(2.75E-2)≈	8.3079E-1(2.40E-2)≈	8.3647E-1(3.97E-2)
Mice	9.4274E-1(1.54E-2)+	9.2745E-1(9.15E-3)+	9.5034E-1(8.29E-3)+	9.3688E-1(1.37E-2)+	9.4082E-1(1.39E-2)+	9.5906E-1(2.76E-3)
Musk1	8.3645E-1(1.89E-2)+	8.4660E-1(1.57E-2)+	8.6009E-1(2.04E-2)+	8.3853E-1(1.77E-2)+	8.2884E-1(2.00E-2)+	8.8227E-1(1.90E-2)
Semeion	8.3694E-1(1.73E-2)+	8.8863E-1(2.16E-2)+	9.0383E-1(1.81E-2)+	8.3056E-1(1.57E-2)+	8.3400E-1(1.98E-2)+	9.6999E-1(1.28E-2)
LSVT	7.9345E-1(4.86E-2)+	7.9127E-1(4.42E-2)+	8.0344E-1(3.83E-2)+	7.6225E-1(3.84E-2)+	7.8222E-1(3.63E-2)+	8.5663E-1(4.14E-2)
Madelon	5.8973E-1(2.15E-2)+	6.4972E-1(3.14E-2)+	5.4590E-1(2.50E-2)+	5.8568E-1(1.86E-2)+	5.7459E-1(4.25E-2)+	9.2011E-1(3.25E-3)
colon	3.5534E-1(5.59E-2)+	4.0823E-1(4.55E-2)+	3.9570E-1(6.20E-2)+	3.5839E-1(4.02E-2)+	3.5404E-1(4.86E-2)+	5.4519E-1(1.99E-3)
SRBCT	5.9123E-1(2.62E-2)+	6.3356E-1(4.39E-2)+	6.4351E-1(5.04E-2)+	5.9868E-1(3.19E-2)+	5.9203E-1(3.63E-2)+	9.2806E-1(7.23E-2)
AR10P	3.3300E-1(2.44E-2)+	3.2874E-1(2.14E-2)+	3.2747E-1(2.70E-2)+	3.3133E-1(2.79E-2)+	3.2439E-1(3.13E-2)+	6.7223E-1(6.82E-2)
lymphoma	4.1382E-1(4.09E-2)+	4.4093E-1(4.51E-2)+	4.3083E-1(4.96E-2)+	4.2947E-1(2.93E-2)+	4.3073E-1(3.01E-2)+	5.8231E-1(6.47E-2)
Leukemia1	5.7995E-1(5.01E-3)+	6.1562E-1(6.42E-3)+	6.3701E-1(4.52E-3)+	5.8782E-1(5.53E-3)+	5.8466E-1(4.53E-3)+	9.4560E-1(4.94E-2)
DLBCL	6.0147E-1(3.17E-2)+	6.2171E-1(3.14E-2)+	6.3108E-1(4.44E-2)+	6.1066E-1(2.88E-2)+	6.0564E-1(3.00E-2)+	9.3239E-1(6.03E-2)
Brain1	3.7500E-1(6.25E-2)+	3.7315E-1(4.32E-2)+	4.1126E-1(7.65E-2)+	3.6763E-1(5.20E-2)+	3.7395E-1(6.12E-2)+	6.5449E-1(1.23E-1)
+ / - / ≈	16/0/2	17/0/1	16/0/2	17/0/1	16/0/2	-
Friedman's rank	4.00	4.11	2.72	4.67	4.39	1.11

the performance of the proposed MFFS algorithm and its competitors. These datasets come from different domains such as medical, financial, signal processing, handwritten digits, and image.

B. Comparison Algorithms

First, five classical or state-of-the-art MOEAs are chosen as competitors of MFFS: 1) NSGA-II [28]; 2) MOEA/D [41]; 3) MOEA/D-AWA [42]; 4) CA-MOEA [46]; and 5) DEA-GNG [47]. The reasons for choosing the above five MOEAs as competitors are based on the following considerations: the environmental selection operator of the multiobjective FS task in MFFS also uses the nondominated sorting and crowding distance as in NSGA-II, while the fitness function of the single-objective FS tasks in MFFS is similar to MOEA/D. Since the Pareto front of the multiobjective FS problem is usually irregular, for example, discontinuous, three MOEAs (i.e., MOEA/D-AWA, CA-MOEA, and DEA-GNG) tailed for handling problems with irregular Pareto fronts are also adopted for comparisons.

Furthermore, MFFS is also compared with the following four state-of-the-art MOEA-based methods that are specifically designed for FS, which have shown their effectiveness in handling multiobjective FS problems: SparseEA [33], SIOM-NSGA-II [48], PMMOEA [34], and DAEA [10].

C. Parameter Settings

The parameter settings of the compared algorithms follow their original papers. The experiments for MOEAs are conducted on the open-source platform PlatEMO [49]. Each MOEA independently runs 30 times on each dataset.

Each classification dataset is randomly divided into a training data subset and a test data subset with the proportions of about 70% and 30% [6], [11], respectively. During the training process, KNN with five-fold cross-validation is utilized to calculate the classification error on the training data subset to avoid the FS bias [50], while the number of

nearest neighbors is set to 5 for the balance between accuracy and efficiency [11], [48]. Note that all the above settings are commonly used in the evolutionary FS community. The nondominated feature subsets obtained at the end of the evolutionary training process are applied to the unseen test set to calculate the classification error rates.

Except for SparseEA, SIOM-NSGA-II, PMMOEA, and DAEA which employ their specific genetic operators, all the other compared algorithms and the proposed MFFS algorithm use the single-point crossover and bit-flip mutation operators to generate offspring solutions, where the crossover probability and mutation probability are set to 1.0 and $1/D$, respectively. The population size is set to be the same as the number of features ($N = D$) but is bounded by 200 ($N = 200$ if $D > 200$) to avoid high computational costs on high-dimensional datasets [48]. For MFFS, there is a parameter l which determines how many consecutive generations of single-target FS tasks should be reinitialized after no improvement. In this article, we empirically set $l = 5$.

Two widely used metrics: 1) inverted generational distance (IGD) [51] and 2) hypervolume (HV) [52], are employed to measure the quality of feature subsets obtained by each MOEA. Due to the true Pareto front of FS in each dataset is usually unknown, to calculate IGD, we first collect the non-dominated solutions obtained by all compared MOEAs on the test set and merge them into a union population. Subsequently, the nondominated solutions in the union population are taken as the “true” Pareto front. For HV, all the objective values obtained by each MOEA are normalized by the ideal and nadir points to the scale $[0, 1]$ in each objective direction, while the ideal and nadir points are set to $(0, 0)$ and $(1.1, 1.1)$, respectively, and the reference point of HV is set to $(1, 1)$ according to the suggestion in [10] and [48].

Wilcoxon's rank-sum test at a 0.05 significance level was conducted to compare the proposed method with each compared algorithm. The symbols +, -, and ≈ represent that MFFS performs significantly better than, significantly worse than, and statistically tied the compared MOEA, respectively.

TABLE II
COMPARATIVE RESULTS OF HV VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY
SPARSEEA, SIOM-NSGA-II, PMMOEA, DAEA, AND MFFS ON THE TEST DATA

Dataset	SparseEA [33]	SIOM-NSGA-II [48]	PMMOEA [34]	DAEA [10]	MFFS
Leaf	5.6360E-1(1.46E-2)+	6.1014E-1(7.42E-3)+	5.7662E-1(1.63E-2)+	5.8419E-1(1.10E-2)+	6.2347E-1(6.38E-3)
German	6.9042E-1(9.21E-3)≈	6.9049E-1(1.02E-2)≈	6.8738E-1(1.22E-2)≈	6.9095E-1(4.61E-3)≈	6.9179E-1(1.13E-4)
WBCD	9.4548E-1(4.51E-3)≈	9.4067E-1(6.98E-3)+	9.4503E-1(2.38E-3)+	9.4459E-1(3.38E-3)+	9.4549E-1(2.38E-3)
Dermatology	8.8894E-1(5.17E-3)≈	8.9258E-1(4.94E-3)≈	8.8863E-1(4.12E-3)≈	8.9031E-1(4.97E-3)≈	8.9082E-1(4.55E-3)
Ionosphere	9.0096E-1(5.81E-3)≈	8.9670E-1(7.68E-3)+	9.0361E-1(9.21E-3)≈	9.0180E-1(7.81E-3)≈	9.0299E-1(7.62E-3)
Sonar	8.3477E-1(2.90E-2)≈	8.2234E-1(3.40E-2)≈	8.2583E-1(3.04E-2)≈	8.3625E-1(2.42E-2)≈	8.3647E-1(3.97E-2)
Mice	9.5848E-1(3.58E-3)≈	9.5871E-1(5.09E-3)≈	9.5872E-1(5.13E-3)≈	9.6111E-1(2.95E-3)−	9.5906E-1(2.76E-3)
Musk1	8.8596E-1(1.71E-2)≈	8.9471E-1(1.61E-2)−	8.7522E-1(2.07E-2)≈	9.0168E-1(1.55E-2)−	8.8227E-1(1.90E-2)
Semeion	9.5893E-1(8.44E-3)+	9.3518E-1(1.35E-2)+	9.5560E-1(1.44E-2)+	9.5879E-1(9.29E-3)+	9.6999E-1(1.28E-2)
LSVT	8.2425E-1(3.52E-2)+	8.5956E-1(5.18E-2)≈	8.1655E-1(5.04E-2)≈	8.6119E-1(3.62E-2)≈	8.5663E-1(4.14E-2)
Madelon	9.1727E-1(4.62E-3)+	8.0486E-1(6.34E-2)+	9.1403E-1(8.60E-3)+	8.8474E-1(1.88E-2)+	9.2011E-1(3.25E-3)
colon	5.4521E-1(2.26E-4)−	4.3905E-1(5.66E-2)+	5.0012E-1(5.45E-2)+	5.3688E-1(3.18E-2)+	5.4519E-1(1.99E-3)
SRBCT	9.2536E-1(2.21E-2)≈	7.6382E-1(3.54E-2)+	9.0736E-1(5.80E-2)≈	9.3683E-1(3.85E-2)≈	9.2806E-1(7.23E-2)
AR10P	6.8405E-1(4.98E-2)≈	4.5238E-1(4.48E-2)+	6.2967E-1(7.71E-2)+	6.7247E-1(4.17E-2)≈	6.7223E-1(6.82E-2)
lymphoma	5.3022E-1(7.40E-2)+	5.2396E-1(5.50E-2)+	5.0934E-1(7.79E-2)+	5.1943E-1(7.05E-2)+	5.8231E-1(6.47E-2)
Leukemia1	9.3445E-1(6.93E-2)≈	7.6723E-1(7.39E-3)+	9.2249E-1(7.70E-2)≈	9.2933E-1(4.50E-2)≈	9.4560E-1(4.94E-2)
DLBCL	9.0691E-1(6.47E-2)≈	7.8870E-1(5.56E-2)+	9.1821E-1(7.21E-2)≈	9.2634E-1(6.69E-2)≈	9.3239E-1(6.03E-2)
Brain1	6.3440E-1(9.11E-2)≈	5.1963E-1(8.81E-2)+	6.2431E-1(1.44E-1)≈	6.6112E-1(7.92E-2)≈	6.5449E-1(1.23E-1)
+ / − / ≈	5/1/2	12/1/5	7/0/11	6/2/10	−
Friedman's rank	3.00	4.00	3.89	2.39	1.72

V. RESULTS AND DISCUSSIONS

A. MFFS Versus Classical and State-of-the-Art MOEAs

The comparison results derived from NSGA-II, MOEA/D, MOEA/D-AWA, CA-MOEA, DEA-GNG, and MFFS in terms of HV [52] are summarized in Table I. At the first glance, the proposed MFFS method can achieve superior performance on most datasets. Specifically, both NSGA-II and MOEA/D-AWA exhibit the best performance on one dataset, and the proposed MFFS method exhibits the best performance on the rest 16 datasets, while the rest three algorithms (MOEA/D, CA-MOEA, and DEA-GNG) do not gain any best result. According to statistical significance tests, the proposed MFFS method is significantly better than NSGA-II, MOEA/D, MOEA/D-AWA, CA-MOEA, and DEA-GNG on 16, 17, 16, 17, and 16 datasets, respectively. Besides, Friedman's test results show that MFFS has the best ranking among the compared methods.

Similar results can be observed in Table S-II (in the supplementary material) in terms of IGD. From Table S-II in the supplementary material, we can see that MFFS is significantly better than the five competitors on 16, 17, 16, 17, and 16 out of the 18 datasets, respectively, which is far greater than the number where the performance of MFFS is significantly worse than its peer algorithms.

B. MFFS Versus State-of-the-Art MOEA-Based FS Methods

The experimental results among MFFS and the four state-of-the-art MOEA-based FS methods in terms of HV values are collected in Table II. It can be seen from Table II that MFFS has the best performance on 9 out of the 18 datasets. For the rest of the nine datasets, SparseEA performs the best on three datasets; SIOM-NSGA-II and PMMOEA achieve the best results on two and one datasets, respectively, and DAEA wins on five datasets. Wilcoxon's rank-sum test results show that MFFS is significantly better than the four state-of-the-art FS competitors on 5, 12, 7, and 6 out of the 18 datasets, respectively. By contrast, the number of datasets that MFFS

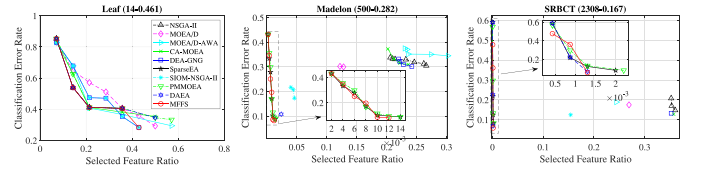


Fig. 4. Distributions of nondominated solutions obtained by each algorithm on test sets in terms of median HV value.

performs significantly worse than its competitors is one, one, zero, and two, respectively. Friedman's test results in Table II show that MFFS ranks first among the five FS algorithms.

Table S-III (in the supplementary material) reports the results in terms of IGD values. It can be observed from Table S-III in the supplementary material that MFFS has the best results on eight datasets. Wilcoxon's rank-sum test confirms that MFFS is significantly better than its four compared methods on 5, 13, 11, and 8 datasets, respectively, which is greater than the number where the performance of MFFS is significantly worse than its peer algorithms.

In summary, the above comparison results suggest that MFFS can achieve encouraging results compared with the four state-of-the-art MOEA-based FS methods.

C. Analysis of the Distribution of Nondominated Feature Subsets

Fig. 4 plots the final nondominated solutions with the median HV obtained by the compared 10 MOEAs on test sets. Due to the page limit, we only present results on the Leaf, Madelon, and SRBCT datasets, which represent the datasets with a small, medium, and large number of features.

From Fig. 4, we can see that the compared five classical and state-of-the-art MOEAs (i.e., NSGA-II, MOEA/D, MOEA/D-AWA, CA-MOEA, and DEA-GNG) cannot obtain better results on medium and high-dimensional datasets, since they not only select much more features but also have worse classification performance than that of MFFS. Although three

TABLE III
AVERAGE MCER OF THE TEN COMPARISON ALGORITHMS

Dataset	NSGA-II	MOEA/D	MOEA/D-AWA	CA-MOEA	DEA-GNG	SparseEA	SIOM-NSGA-II	PMMOEA	DAEA	MFFS
Leaf	37.37%+	31.79%+	29.78%≈	34.79%+	28.94%≈	38.66%+	28.81%≈	35.94%+	34.70%+	27.76%
German	31.30%≈	33.15%+	31.67%≈	32.56%≈	31.78%≈	30.91%≈	30.91%≈	31.28%≈	30.98%≈	30.91%
WBCD	3.177%+	4.702%+	3.898%+	3.347%≈	3.369%+	2.336%≈	2.916%≈	2.389%≈	2.442%≈	2.336%
Dermatology	3.742%≈	5.369%+	4.132%≈	3.937%≈	4.247%≈	4.093%≈	3.887%≈	3.934%≈	4.054%≈	3.890%
Ionosphere	10.01%+	10.52%+	9.216%+	9.984%+	9.385%+	7.984%≈	8.519%+	7.644%≈	7.866%≈	7.724%
Sonar	15.29%≈	15.99%≈	13.86%—	15.24%≈	14.99%≈	15.17%≈	17.26%+	16.69%+	15.15%≈	15.50%
Mice	1.332%≈	2.149%+	1.369%≈	1.518%≈	1.474%≈	1.386%≈	1.462%≈	1.406%≈	1.104%—	1.362%
Musk1	9.722%≈	12.17%≈	11.30%≈	9.132%—	10.37%≈	11.25%≈	10.28%≈	12.53%≈	9.196%—	11.55%
Semeion	2.645%+	7.392%+	3.101%+	2.302%≈	2.571%+	3.186%+	2.917%+	3.853%+	2.517%+	1.892%
LSVT	13.46%—	17.38%+	15.69%≈	15.06%≈	15.10%≈	18.89%+	14.15%≈	19.83%+	14.50%≈	15.33%
Madelon	30.51%+	29.05%+	33.07%+	30.27%+	31.23%+	8.759%≈	17.59%+	9.112%≈	10.72%≈	8.447%
colon	55.61%+	54.17%+	57.31%+	54.64%+	55.42%+	50.00%≈	55.44%+	54.96%+	50.92%≈	50.00%
SRBCT	13.69%+	16.40%+	16.13%+	13.18%+	14.61%+	8.144%≈	12.20%+	10.12%+	6.876%≈	7.846%
AR10P	54.36%+	59.28%+	59.11%+	55.14%+	56.00%+	34.70%≈	51.10%+	40.68%≈	35.95%≈	35.98%
lymphoma	39.37%≈	40.13%≈	41.26%≈	37.05%—	36.67%—	51.62%+	41.47%—	53.92%+	52.81%+	45.80%
Leukemia1	8.693%+	8.924%+	9.118%+	8.740%+	9.023%+	7.186%+	8.534%+	8.497%+	7.743%+	5.801%
DLBCL	4.960%—	7.405%≈	6.242%≈	3.730%—	4.762%—	10.22%+	5.744%—	8.970%≈	8.079%≈	7.381%
Brain1	44.05%+	48.38%+	41.58%≈	45.65%+	44.03%≈	40.19%≈	41.24%≈	41.30%≈	37.24%≈	37.88%
+ / - / ≈	10/2/6	14/0/4	8/1/9	8/3/7	9/2/7	6/0/12	8/2/8	8/0/10	4/2/12	—
Friedman's rank	5.44	8.28	6.89	5.61	6.11	4.83	4.83	6.11	3.67	3.22

of the above five algorithms are designed for solving problems with irregular Pareto front, they cannot effectively solve the multiobjective FS problems. This is because they do not consider the characteristics of the multiobjective FS problem, which is very different from numeric continuous optimization benchmark problems. Specifically, they can not guarantee the sparsity of the obtained solutions, that is, reducing the number of selected features. Thus, the number of selected features in the nondominated feature subsets obtained by these five algorithms is much larger than that of the proposed algorithm. In addition, they do not handle the duplicated solutions during the evolutionary process, which can lead to the poor diversity of the population. By contrast, thanks to the single-objective FS tasks in the early stage of evolution, the proposed MFFS algorithm can ensure the sparsity of solutions, which suggests it can find feature subsets with a small number of selected features and lower classification error rates. In addition, benefiting from the proposed diversity maintenance mechanism, MFFS can evolve a more diverse nondominated solution set than that of these five compared algorithms.

In comparison with four state-of-the-art MOEA-based FS methods (i.e., SparseEA, SIOM-NSGA-II, PMMOEA, and DAEA), it can be observed from Fig. 4 that the proposed MFFS algorithm generally has better convergence since it can provide nondominated solutions with lower classification error rates and a smaller number of selected features. Furthermore, the nondominated solutions produced by the proposed MFFS algorithm are generally more diverse.

D. Classification Performance Comparison

In this section, we compare the performance of different algorithms on another performance metric: the minimal classification error rate (in short called MCER) [10], which represents the classification error rate on the test data for the feature subset that has the minimum classification error rate on the training data. Note that among a set of candidate feature subsets obtained by an MOEA, the one with the minimum classification error rate is preferred by decision makers since the classification performance is more important than the

number of selected features. Table III summarizes the average MCER results on the test data over the 30 runs. It can be observed that MFFS can have the best MCER values on most of the datasets. In terms of Friedman's test results, MFFS achieves the first rank among the ten compared algorithms, followed by DAEA.

E. Number of Selected Features Comparison

In this section, we compare the proposed MFFS method with its counterpart methods in terms of the FN metric, which represents the number of selected features by the feature subset with the minimum classification error rate.

Table S-IV of the supplementary material lists the average FN results for the ten comparison algorithms. It can be observed from the table that MOEA/D usually selects the minimum number of features on the small datasets. On the medium and large datasets, NSGA-II, MOEA/D, MOEA/D-AWA, CA-MOEA, DEA-GNG, and SIOM-NSGA-II select a much larger number of features than SparseEA, PMMOEA, DAEA, and MFFS. Among SparseEA, PMMOEA, DAEA, and MFFS, the number of features selected by SparseEA and PMMOEA is generally smaller than that of DAEA and MFFS. This is because SparseEA and PMMOEA belong to the sparse-based MOEA, which focuses more on the number of selected features than the classification performance. However, through the classification performance in Table IV of the main paper, it can be seen that SparseEA and PMMOEA generally obtain worse classification performance than that of DAEA and MFFS.

VI. FURTHER ANALYSIS

A. Effectiveness of the Multiform Framework

In order to study how different forms of FS tasks work, and the advantages of multiform search over other baseline methods, in this section, we examine the movement of the populations of MFFS and NSGA-II at different search stages. Note that both two algorithms start from the same initial population. Fig. 5 considers Musk1 as an example to illustrate the positions of different populations in the

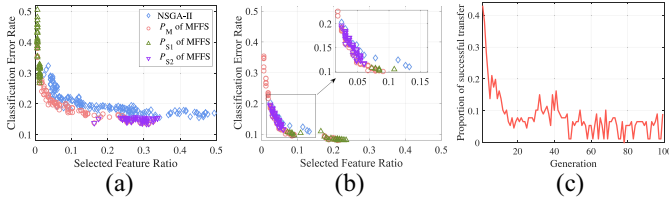


Fig. 5. Evolutionary processes of different populations and the proportion of successful transfer from single-objective tasks to the multiobjective FS task. (a) In the 20th generation. (b) In the 80th generation. (c) Successful transfer proportion.

objective space in the early and later stages of evolution, respectively.

From Fig. 5(a), we can see that in the early stage of evolution, the populations of $\mathcal{P}_{S_1}$ and $\mathcal{P}_{S_2}$ in MFFS for single-objective FS tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$ can search solutions with better objective values in terms of selected feature ratio and classification error rate, respectively. With the help of $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$, the population $\mathcal{P}_M$ of task $\mathcal{T}_M$ has better convergence than that of NSGA-II since it can dominate most solutions obtained by NSGA-II. In the late stage of evolution, from Fig. 5(b), it is obvious that the population $\mathcal{P}_M$ has better diversity than the population obtained by NSGA-II. The population of NSGA-II loses its diversity and only focuses on a small region of the objective space for search.

Fig. 5(c) plots the proportion of successful transfer from tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$ to the task $\mathcal{T}_M$, which is equal to the proportion of offspring solutions in tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$ that succeed as the parent population of $\mathcal{P}_M$ after the environmental selection operation of $\mathcal{T}_M$ (line 12 of Algorithm 1). As can be seen from Fig. 5(c), during the course of evolution, the solutions in single-objective FS tasks can be continuously transferred to the multiobjective FS task.

The above results validate the effectiveness of the complementary populations using multiform. The search on the original multiobjective FS task can leverage the useful information from the single-objective FS tasks to maintain a tradeoff between diversity and convergence. Meanwhile, the search on the constructed single-objective FS tasks can extensively exploit the specific directions defined by direction vectors and also use the promising solutions around their direction vectors from $\mathcal{P}_M$.

B. Effectiveness of the Diversity Maintenance Mechanism

In this article, the diversity maintenance mechanism in MFFS is used to handle duplicated feature subsets. To validate its effectiveness, we implement a variant of NSGA-II combined with the proposed diversity maintenance mechanism, called NSGA-II-DM. Specifically, in NSGA-II-DM, the environmental selection operator in the original NSGA-II is replaced with the environmental selection operator of MFFS (Algorithm 3).

Fig. 6 plots the final nondominated solutions obtained by NSGA-II-DM and NSGA-II on test sets. We can see that on the small and medium datasets, Leaf and Musk, the nondominated solutions obtained by NSGA-II-DM are more diverse than that of NSGA-II, especially in terms of classification error rate. In particular, four out of the seven and three out of the seven

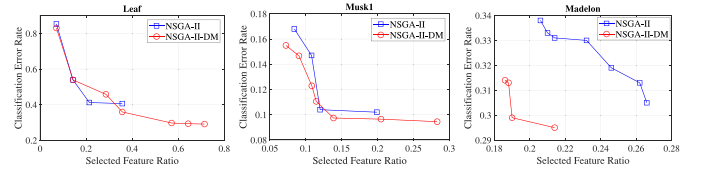


Fig. 6. Distributions of nondominated solutions obtained by NSGA-II and NSGA-II-DM on test sets in terms of median HV value.

solutions obtained by NSGA-II-DM have smaller classification error rates than all solutions obtained by NSGA-II on Leaf and Musk, respectively. On the large dataset of Madelon, the nondominated solution set obtained by NSGA-II-DM has much better convergence than that of NSGA-II as they can dominate the solutions obtained by NSGA-II in terms of the classification error rate and the selected feature ratio. The detailed comparison results between NSGA-II and NSGA-II-DM in Table S-V of the supplementary material also validate the effectiveness of the proposed diversity maintenance mechanism in handling duplicated feature subsets.

C. Comparison on Training Time

Table IV lists the average training time consumption of all compared MOEAs on each dataset over the 30 runs. In general, it can be found that the proposed MFFS algorithm consumes relatively less time on most datasets, and the least training time on 8 out of the 18 datasets.

For datasets with a small number of features, that is, German, WBCD, Ionosphere, and Mice, the classical MOEAs (NSGA-II, MOEA/D, and MOEA/D-AWA) consume the least training time. As the number of features increases, the superiority of MFFS in less training time is gradually demonstrated compared with the five classical or state-of-the-art MOEAs (i.e., NSGA-II, MOEA/D, MOEA/D-AWA, CA-MOEA, and DEA-GNG). The reason is that MFFS is able to search for solutions with a small number of selected features. Since the computational cost of the classification process decreases as the number of selected features decreases, the proposed MFFS algorithm consumes less training time on high-dimensional datasets.

Among the five state-of-the-art MOEA-based FS methods (i.e., SparseEA, SIOM-NSGA-II, PMMOEA, DAEA, and MFFS), the proposed MFFS algorithm has advantages in training time on small and medium-sized datasets (e.g., $D \leq 500$). For datasets with more than 2000 features, PMMOEA consumes a much shorter time than the others on most datasets. Nevertheless, the results show that MFFS can still spend the second least amount of training time on the high-dimensional datasets. The reason can be attributed to the fact that the initial single-objective FS task $\mathcal{T}_{S_1}$ in MFFS mainly focuses on reducing the number of selected features and selecting only a small number of strongly relevant features.

Regarding the scalability of the ten comparison algorithms, NSGA-II, MOEA/D, MOEA/D-AWA, CA-MOEA, DEA-GNG, and SIOM-NSGA-II do not scale well to high-dimensional datasets as they always select a large number of features and consume a long training time. Of the remaining four methods, PMMOEA seems to have the best scalability because it spends the least amount of training time on

TABLE IV
AVERAGE TRAINING TIME OF THE TEN COMPARISON ALGORITHMS (UNIT: SECOND)

Dataset	NSGA-II	MOEA/D	MOEA/D-AWA	CA-MOEA	DEA-GNG	SparseEA	SIOM-NSGA-II	PMMOEA	DAEA	MFFS
Leaf	5.161E+0≈	6.862E+0+	6.554E+0≈	7.312E+0+	8.285E+0+	5.626E+0≈	7.484E+0+	9.286E+0+	5.904E+0≈	4.988E+0
German	5.073E+1≈	5.266E+1≈	4.961E+1≈	5.456E+1≈	5.273E+1≈	5.334E+1≈	5.164E+1≈	5.974E+1≈	5.770E+1≈	5.303E+1
WBCD	1.865E+1≈	1.871E+1≈	1.912E+1≈	2.373E+1≈	1.876E+1≈	2.479E+1≈	2.297E+1≈	2.667E+1≈	2.713E+1+	2.109E+1
Dermatology	1.137E+1≈	1.211E+1≈	1.224E+1≈	1.184E+1≈	1.148E+1≈	1.412E+1+	1.415E+1+	1.433E+1+	1.508E+1+	1.087E+1
Ionosphere	1.033E+1≈	1.154E+1≈	1.241E+1≈	1.118E+1≈	1.041E+1≈	1.693E+1+	1.527E+1+	1.904E+1+	1.705E+1+	1.248E+1
Sonar	9.137E+0≈	1.250E+1+	1.376E+1+	1.170E+1≈	1.210E+1+	2.210E+1+	2.831E+1+	1.340E+1+	2.229E+1+	8.587E+0
Mice	2.306E+2+	1.961E+2≈	2.085E+2≈	2.041E+2≈	2.319E+2+	2.303E+2+	2.078E+2≈	2.108E+2≈	2.275E+2≈	1.980E+2
Musk1	1.231E+2+	1.216E+2+	1.525E+2+	1.365E+2+	1.374E+2+	1.988E+2+	2.436E+2+	1.166E+2≈	2.276E+2+	9.827E+1
Semeion	2.483E+3+	1.774E+3+	2.236E+3+	2.588E+3+	2.547E+3+	1.970E+3+	1.950E+3+	1.646E+3+	2.504E+3+	1.366E+3
LSVT	3.379E+1+	3.739E+1+	4.065E+1+	3.159E+1+	3.371E+1+	1.837E+2+	2.091E+2+	2.858E+1≈	2.008E+2+	2.397E+1
Madelon	1.0531E+4+	9.3882E+3+	1.2061E+4+	1.1153E+4+	1.0482E+4+	3.7808E+3≈	1.3647E+3-	3.3494E+3≈	8.2012E+3+	3.8672E+3
colon	8.936E+1+	8.028E+1+	8.784E+1+	9.783E+1+	9.962E+1+	6.463E+1≈	2.838E+2+	3.396E+1-	2.434E+2+	6.322E+1
SRBCT	1.602E+2+	1.653E+2+	1.563E+2+	1.675E+2+	1.651E+2+	7.300E+1-	3.549E+2+	5.256E+1-	1.018E+2≈	1.084E+2
AR10P	2.911E+2+	3.250E+2+	3.582E+2+	2.969E+2+	2.872E+2+	3.225E+2+	4.302E+2+	8.445E+1-	1.154E+2-	1.746E+2
lymphoma	3.606E+2+	3.281E+2+	3.450E+2+	4.305E+2+	3.417E+2+	3.751E+2+	6.480E+2+	3.129E+2+	4.837E+2+	2.374E+2
Leukemia1	3.586E+2+	3.306E+2+	3.306E+2+	3.628E+2+	3.394E+2+	2.902E+2≈	6.806E+2+	3.041E+2≈	6.174E+2+	2.715E+2
DLBCL	3.923E+2+	3.705E+2+	3.397E+2+	4.769E+2+	4.470E+2+	2.923E+2≈	4.533E+2+	7.293E+1-	6.181E+2+	2.880E+2
Brain1	5.423E+2≈	4.647E+2-	6.060E+2+	5.078E+2+	5.224E+2+	3.008E+2-	4.703E+2-	8.754E+1-	7.395E+2+	5.308E+2
+/-/≈	11/0/7	12/1/5	12/0/6	12/0/6	14/0/4	9/2/7	13/2/3	6/5/7	13/1/4	-
Friedman's rank	4.78	4.67	5.78	6.28	5.72	5.50	7.39	4.33	7.83	2.72

high-dimensional datasets, but its classification performance is not as good as DAEA and MFFS. DAEA and the proposed MFFS method exhibit the best performance in terms of both HV and MCER metrics. However, of these two methods, DAEA incurs a computational complexity of $O(DN^2)$, which prevents it from scaling well to datasets containing tens of thousands of features. By contrast, as discussed in Section III-E, MFFS only yields the computational complexity of $\max\{O(DN), O(N^2)\}$, which suggests its relatively good scalability for high-dimensional data.

To sum up, compared with nine counterpart FS methods, the proposed MFFS method can consume a relatively shorter training time to obtain a set of promising feature subsets in most cases.

D. Robustness Investigation in Noisy Environment

To further assess the performance of MFFS, another experiment in a noisy environment is conducted, which artificially generates noises and includes them in the original dataset. To be specific, as suggested by [53], we manually generate three levels of noises for each dataset: we randomly select about 10%, 20%, and 30% features, respectively, and replace them with random values (between the minimum and maximum value of each feature). Four real-world high-dimensional medical datasets (lymphoma, Leukemia1, DLBCL, and Brain1) that have more than 4000 features are selected to test the performance of each compared FS method.

Tables S-VI–S-VII of the supplementary material record the average HV and MCER values obtained by ten algorithms at 10%, 20%, and 30% level noise, respectively. Through these tables, we have the following observations and discussions.

- 1) In comparison with five classical and state-of-the-art MOEAs (i.e., NSGA-II, MOEA/D, MOEA/D-AWA, CA-MOEA, and DEA-GNG), MFFS is significantly better than these five methods in terms of the HV metric. This is because MFFS can obtain a set of tradeoff feature subsets with a low classification error rate and a small number of selected features, thus contributing a larger HV value. In terms of the MCER metric, MFFS can still

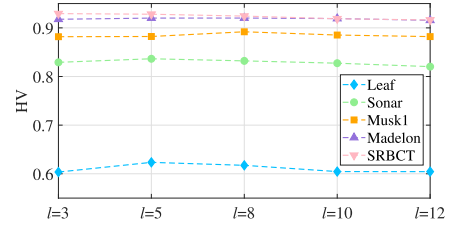


Fig. 7. HV values obtained by different l settings.

be significantly better than the compared algorithms on most of the datasets, while is significantly worse than these five competitors only on one dataset.

- 2) In comparison with four state-of-the-art MOEA-based FS methods (i.e., SparseEA, SIOM-NSGA-II, PMMOEA, and DAEA), MFFS can still have better performance at different noise levels. Both DAEA and MFFS can achieve the best results according to the Wilcoxon test results. In terms of Friedman's rank results, MFFS can rank first in both the HV and MCER metrics, which also shows the robustness of MFFS on the noise environment.

E. Parameter Sensitive Analysis

For the proposed MFFS method, there is a parameter l to determine when the single-objective FS tasks need to be reinitialized. Intuitively, a larger l value allows single-objective FS tasks to extensively exploit the specific directions or areas defined by direction vectors λ_{S_1} and λ_{S_2} , but it can result in a waste of computing resources. By contrast, a smaller l value enables single-objective FS tasks to explore more areas, but it can lead to insufficient search. To test the effect of l on the performance of MFFS, we examine five different values of l : 3, 5, 8, 10, and 12 on the Leaf, Sonar, Musk1, Madelon, and SRBCT datasets, respectively.

The comparison results are summarized in Fig. 7. Among the five l values, $l = 5$ and $l = 8$ show the best overall performance. As a result, l between 5 and 8 is the recommended setting.

F. Effectiveness of the Search Direction Adjustment Strategy

To verify the effectiveness of the search direction adjustment strategy, we conduct an experiment that compares the performance of MFFS without and with the search direction adjustment strategy. Specifically, for MFFS without the search direction adjustment strategy (denoted as MFFS-w/oSDA), the direction vectors λ_{S_1} and λ_{S_2} for single-objective FS tasks $\mathcal{T}_{S_1}$ and $\mathcal{T}_{S_2}$ always keep unchanged [$\lambda_{S_1} = (0.99, 0.01)$ and $\lambda_{S_2} = (0.01, 0.99)$] through the evolutionary learning process. This can test whether the proposed direction adjustment strategy can compensate the diversity loss of the main population of task $\mathcal{T}_M$. We use the HV metric to compare MFFS-w/oSDA and MFFS, since HV can measure both the diversity and convergence ability of a solution set in the objective space.

Table S-VIII of the supplementary material lists the average results obtained by MFFS-w/oSDA and MFFS in terms of HV and MCER metrics, respectively. It can be seen from the table, equipped with the proposed search direction adjustment strategy, MFFS outperforms the MFFS-w/oSDA on most datasets, which suggests the effectiveness of the proposed direction adjustment strategy.

VII. CONCLUSION

This article proposes a multiform FS method, called MFFS, which has made the first attempt to handle a multiobjective FS task together with its associated single-objective FS tasks in a multitask environment. Benefiting from the unique search provided by the constructed single-objective FS tasks, useful knowledge can be continuously provided to the multiobjective FS task, thereby facilitating improving the performance of the multiobjective FS task. The effectiveness and efficiency of MFFS have been verified by comparing it with nine MOEAs on 18 classification datasets.

For future work, it is interesting to investigate other forms of FS tasks, that is, constructing auxiliary tasks by imposing constraints on objectives to focus more on the classification performance. In addition, one shortcoming of MFFS is that it selects a few more features than some state-of-the-art MOEA-based FS methods. Therefore, it is worthwhile designing additional dimension-reduction strategies to further reduce the number of selected features in the future. In addition, it is also worth doing some theoretical work [54] to show the benefits of multiform.

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